MM/PBSA tutorials for SARS-CoV-2 Mpro in complex with inhibitor N3 by MolAICal

Qifeng Bai
Email: molaical@yeah.net
Homepage: https://molaical.github.io or https://molaical.gitlab.io
School of Basic Medical Sciences
Lanzhou University
Lanzhou, Gansu 730000, P. R. China

1. Introduction

In this tutorial, the MolAICal (https://molaical.github.io) is used to calculate the MM/PBSA between ligand N3 and SARS-CoV-2 Mpro based on molecular dynamics (MD) simulated results by NAMD. This tutorial is just a demo. To save running and storage space, only 25 frames of MD simulated trajectories of SARS-CoV-2 Mpro in complex with N3 are selected for this tutorial. This tutorial can **not only** be used to calculate MM/PBSA of protein-ligand **but also** MM/PBSA of protein-peptide, protein-protein, DNA-ligand, DNA-protein, protein-DNA, protein-RNA, and any complex based on the run MD simulations. It just replaces the protein and ligand of this tutorial with the appointed objects. For example, **protein** in this tutorial is substituted for **DNA**, and **ligand** in this tutorial is substituted for **peptide** **based on the same running command arguments of this tutorial**.

2. Materials

2.1. Software requirement

1) MolAICal: https://molaical.github.io
2) NAMD: https://www.ks.uiuc.edu/Research/namd/

(**Notice:** This tutorial can be run by **NAMD 2.x, 3.x, or a higher version.** For example, the command “**namd3**” substitutes for the command “**namd2**” in this tutorial **if you use NAMD 3.x version**. For a higher version of NAMD, users can use a similar replacement way as the previous example.)

3) Carma: https://github.com/glykos/carma or https://utopia.duth.gr/~glykos/Carma.html
4) APBS (https://apbs.readthedocs.io) or DelPhi (http://compbio.clemson.edu/lab/delphisw)

2.2. Example files

1) All the necessary tutorial files are downloaded from:
https://github.com/MolAICal/tutorials/tree/master/020-mmpbsa

3. Procedure

Part I. Problem solution

Problem: Some users meet the positive value problem when calculating MM/GBSA, which may happen in MM/PBSA calculations:
https://www.ks.uiuc.edu/Research/namd/mailing_list/namd-l.2020-2021/1295.html

Issues: ΔG binding is positive. During the MD simulations, one independent molecule may go into another image. So it seems no interactions with the other molecule if it only depends on distances. Actually, they still have the interaction because of periodic boundary conditions. If the users' research subject does not involve multiple molecules entering different images, **this section can be skipped, and** users may proceed directly **to Part II.**

Possible solutions: it must check whether the ligand is always in the same periodic boundary with

the receptor along the entire simulated time. If not, please use the VMD PBC to wrap the ligand into the receptor along simulated time. More detailed:
https://www.ks.uiuc.edu/Research/vmd/script_library/scripts/pbcwrap/
https://www.ks.uiuc.edu/Research/vmd/plugins/pbctools/

I supply two commands to run on the VMD Tk Console:
%> pbc wrap -centersel protein -center com -compound chain -all

Note: checking the complex in VMD, if the above command can wrap the complex together. It can skip this command below (pbc unwrap):

%> pbc unwrap -sel "not (water or ions)" -all

Running above one or two commands can wrap the ligand into the receptor over the simulated time. Saving wrap or unwrap trajectories into a DCD file using the below command:
%> set all [atomselect top all]
%> animate write dcd mpro.dcd sel $all beg 0 end 24 waitfor all

Here, all atoms are selected. Users can select the wanted part. Our tutorial trajectories only have 25 frames. So the beginning is 0 and the end is 24. "mpro.dcd" is just a name. And then starting the following steps.

Option (Wrapping operation without GUI)

If users have wrapped all complex or have no requirement for the DCD wrapping operation, or VMD and NAMD have already been configured within the Linux version of the MolAICal container, users can skip this option directly.

Users may execute and test the aforementioned commands within either Windows or Linux graphical desktop environments. However, **when operating in environments lacking graphical interfaces, command-line input becomes necessary.** To facilitate external program execution, MolAICal provides a persistent configuration interface. The following command establishes the name and path for VMD (Visual Molecular Dynamics), **requiring only a single configuration that remains available for subsequent use without repetition**.

To address the library dependency issues in Linux, **the Linux version of MolAICal (Not for Windows version)** adopts a container-based approach. If **external programs need** to be invoked, it is **recommended** to install them **within** the MolAICal **container**. If **no external** programs are required, this step can be **ignored**. This tutorial requires the external programs NAMD and VMD, and the steps are as follows:

a) First of all, copy the files to the MolAICal container.

```
****** 1.   Go to the container file system, it will enter the "/root" ******
#> molaical.exe -eset shell in

****** 2.   The first part of 'cp' is from the local machine, and the second part is in the container; ******
****** The VMD and NAMD packages can be moved into the container by the command below ******
#> cp /home/user/<local machine file> /root/soft

****** 3.   Exit container file system ******
#> exit
```

b) Secondly, enter the container virtual environment of MolAICal:

```
#> molaical.exe -eset sys run molaical
```

Note: 'molaical' is the container name.

c) Installing software in the container virtual environment of MolAICal is the same as installing it on the host local machine. Here, we take the installation of VMD and NAMD as examples:

1) For NAMD:

Extract the NAMD files, assuming the extracted folder is named namdcpu. Then specify its path to MolAICal using the following command:

```
#> molaical.exe -call set -n NAMD -p "/root/soft/namdcpu/namd3"
```

Note: Please **replace** the above paths for VMD and NAMD with the **actual paths** on the users' system. The string "VMD" and "NAMD" (case insensitive) after "-n" is fixed for the paths of VMD and NAMD.

2) For VMD:

Following the steps below:

- ✧ Extract the VMD files:

```
#> tar -xzvf vmd-xxx.tar.gz
```

Note: Please **replace** the above paths for VMD and NAMD with the **actual paths** on the users' system.

- ✧ Modify the install path in a file named "**configure**" in the decompressed folder of VMD:

```
The default: $install_bin_dir="/usr/local/bin"; modify it to →
$install_bin_dir="/root/soft/vmd193/bin";

The default: $install_library_dir="/usr/local/lib/$install_name"; modify it to →
$install_library_dir="/root/soft/vmd193/lib/$install_name";
```

- ✧ Install VMD:

```
#> cd vmd-xx
#> ./configure LINUXAMD64
```

```
#> cd src
#> make install
```

Note: Please run **./configure**, and select the right types for the used computers. Here, it is "LINUXAMD64".

✧ Then specify its path to MolAICal using the following command:

```
#> molaical.exe -call set -n VMD -p "/root/soft/vmd193/bin/vmd"
```

So far, all the installations and configurations of NAMD and VMD are complete in the MolAICal container.

✧ **Remember to use the "exit" command to quit the MolAICal virtual environment and return to the local computer for computation (primarily to avoid file-copying steps; execution within the container is also possible but requires copying data from the local computer to the MolAICal container).**

```
#> exit
```

To prevent the loss of installed programs (e.g., VMD and NAMD) when rebuilding the MolAICal image, refer to step 3 in **Appendix 1**.

Then, run the arguments of the external program for DCD wrapping operation without GUI:

```
#> molaical.exe -call run -n vmd -i -dispdev text -e pbcwrap.vmd
```

Note:

1) In MolAICal's environment variable configuration, the parameter name following '-n' (e.g., '-n VMD') must match the corresponding runtime parameter (e.g., '-n vmd') in terms of spelling, though case sensitivity is not enforced. As shown in the two commands above, the parameter names following '-n' must be consistent.
2) The '-i' flag is followed by VMD runtime arguments. '-i' must be placed after other arguments. Here, simply replace pbcwrap.vmd with the users' own script file, though users may optionally use this tutorial-provided script pbcwrap.vmd.

"pbcwrap.vmd" is shown as an example below (it can be modified according to the purpose or assay results of users):

```
package require pbctools
mol new mpro.psf
mol addfile mpro.dcd waitfor all
pbc wrap -centersel protein -center com -compound chain -all
set all [atomselect top all]
animate write dcd mpro.dcd sel $all beg 0 end 24 waitfor all
quit
```

Part II. MM/PBSA calculation based on NAMD

Assuming users have configured the internal commands in MolAICal for VMD and NAMD. The above instructions are for installing external programs in the Linux version of the MolAICal container. For the **Windows version of MolAICal**, users can replace the paths in the command below with the actual system paths and then run it directly:

```
#> molaical.exe -call set -n VMD -p "C:\Program Files (x86)\University of Illinois\VMD\vmd.exe"
#> molaical.exe -call set -n namd -p "d:\namd2\namd2.exe"
#> molaical.exe -call set -n delphi -p "E:/delphi/delphicpp_v8.4.3_windows_x64.exe"
```

Note: Please **replace** the above paths for VMD and NAMD with the **actual paths** on the users' system. When running MMPB/GB/SA functionality, and setting environment variables using MolAICal (e.g., #> molaical.exe -call set -n VMD -p "path"), the variable names in MMPB/GB/SA corresponding to '-n' are fixed: 'vmd', 'namd', and 'delphi' (case insensitive).

- ✧ DelPhi is a binary file and requires executable permission on Linux systems: "chmod +x delphicpp_v8.4.3_windows_x64.exe". Therefore, users can refer to the NAMD installation steps in Appendix 1 or the above steps.
- ✧ Please note that the configuration of MolAICal differs between the Windows and Linux versions: the Linux version utilizes the udocker container approach and requires setup within the container, whereas the Windows version does not.

3.1. MM/PBSA calculation

Firstly, I assume the equilibrium trajectory named "mpro.dcd" has been run by NAMD. Receptor and ligand have been put into the same periodic boundary in "mpro.dcd". Users can replace "mpro.dcd" with their own trajectory. Go to the tutorial directory:

#> cd 020-mmpbsa

And run the below command that can quickly calculate MM/PBSA:

#> molaical.exe -mmpbgbsa -f mmpbsa_apbs.vmd

- ✧ It can modify the arguments in the file 'mmpbsa_apbs.vmd', or modify the arguments by inputting '-i' argument. For more detailed information, please check the MolAICal manual.
- ✧ **Option:** here, the input files such as 'mpro.dcd' and 'mpro.psf' in the input file 'mmpbsa_apbs.vmd' consist of protein, ligand, ions, water, etc, which may occupy much computer memory. To save memory for computation, users can use the command below to only extract the trajectory of the receptor in complex with the ligand:

```
#> molaical.exe -call run -n vmd -c stripDCD -i -dispdev text -psf "mpro.psf" -args protein,or,resname,LIG "mpro.dcd" "complex" mpro.psf mpro.pdb
```

Note: "complex" is the prefix of the output files. The mpro.psf and mpro.pdb are referred to generate the files that have the prefix "complex". It will generate 'complex.psf', 'complex.pdb', and 'complex.dcd', which can be used to substitute for 'mpro.psf', 'mpro.pdb', and 'mpro.dcd' in the input file 'mmpbsa_apbs.vmd' to save loading memory.

Run the above command, and the output results of MM/PBSA and the binding free energy △G are as follows:

Delta:					
BOND:	25	-0.0000		0.0000	0.0001
ANGLE:	25	-0.0000		0.0000	0.0000
DIHED:	25	-0.0000		0.0000	0.0000
IMPRP:	25	0.0000		0.0000	0.0001
KINETIC:	25	14.0776		0.0000	0.0000
Elec:	25	-30.3701		1.3023	6.5114
Vdw:	25	-52.1546		1.0534	5.2670
PB:	25	23.7231		2.3847	11.9234
SA:	25	-7.6990		0.1004	0.5022
Gas:	25	-82.5247		1.7298	8.6491
Sol:	25	16.0241		2.3622	11.8111
Pol:	25	-6.6470		2.6141	13.0705
Npol:	25	-59.8536		1.1223	5.6116
G binding:	**25**	**-66.5006**	+/-	**2.7810**	**13.9052**

Note: all energy values are in kcal/mol. Results may vary across different operating systems or NAMD versions, but if two decimal places are retained, the results will be identical or very similar.

Input files and result analysis:

Open 'mmpbsa_apbs.vmd', please see two arguments: "-poisson_boltzmann" and "-pb_executable".

- If users do not set the argument "-pb_executable", MolAICal will automatically call the inner APBS to calculate PB energy.
- If "-poisson_boltzmann" is set to 2, it will invoke ABPS to calculate PB energy. If "-poisson_boltzmann" is set to 1, it will invoke DelPhi to calculate PB energy. If "-poisson_boltzmann" is set to 0, it will carry out MM/GBSA.

  If users set "-poisson_boltzmann" to 1 and "-pb_executable" to the path of DelPhi, which is named 'mmpbsa_delphi.vmd' in this tutorial, run the command below again, it will call DelPhi to calculate PB energy:

  ```
  #> molaical.exe -mmpbgbsa -f mmpbsa_delphi.vmd
  ```

- For the file 'mmpbsa_apbs_create_file.vmd' or 'mmpbsa_delphi_create_file.vmd', it has two arguments than 'mmpbsa_apbs.vmd' or 'mmpbsa_delphi.vmd':
  - "-namd_config_file   input.namd \".
  - "-pb_gb_file   input.apbs \"   **or**   "-pb_gb_file   input.delphi \"
  - "-complex_selection", "-receptor_selection", and "-ligand_selection" are used to select the molecular part of the complex, receptor, and ligand that is like the command "**atomselect**" in the VMD Tk console.
  - "-sa_input_apbs" is responsible for customizing the APBS input file for SASA calculation. Users can refer to the template file named "input_sa.apbs" in the material files. Just put the "-sa_input_apbs   input_sa.apbs \" a new line of the input file *.vmd, such as 'mmpbsa_apbs.vmd' or 'mmpbsa_delphi.vmd'

  This allows users to customize input parameters of NAMD in the file "input.namd" and input parameters of APBS in the file "input.apbs", or input parameters of DelPhi in the file "input.delphi". If running the command below:

```
#> molaical.exe -mmpbgbsa -f mmpbsa_apbs_create_file.vmd
or
#> molaical.exe -mmpbgbsa -f mmpbsa_delphi_create_file.vmd
```

If setting the same parameters in the custom files "input.namd" and "input.apbs" as the inner parameters of MolAICal, it will get the same results.

- As shown in the above result table, the polar contribution is Pol = Elec + PB, while the nonpolar contribution is Npol = Vdw + SA.

The detailed arguments for 'mmpbsa_apbs_create_file.vmd', 'mmpbsa_delphi_create_file.vmd', 'mmpbsa_apbs.vmd' or 'mmpbsa_delphi.vmd' are as follows:

```
usage: mmpbgbsa
-f <arg>                 The path of the input file for MM/PB/GB/SA calculation.
-h                       Print help information.
-i                       The command arguments of the program mmpbgbsa. It includes all command-line
                         arguments of external programs, but the '-i' parameter must be the last one specified,
                         while all other identifiers (e.g., '-f', etc.) must come before '-i'.
-mmpbgbsa                MM/PB/PBSA calculation.
-n <arg>                 The name of the executed program VMD, the default value is 'vmd'.
-o <arg>                 Whether to print the output of the program.
-p <arg>                 The path of the program VMD. The MolAICal environment setup command can be
                         used to configure it once, eliminating the need for tedious path input later.

Usage: -mmgbpbsa -f xxx.vmd -i -system_topology top_file -system_coords coords_file -trajectory_files
filename [-args...]
Mandatory arguments:
-system_topology         <topology filename>
-trajectory_files        <trajectory filename>
-system_coords           <coordinates filename>
Optional arguments:
-namd_config_file        <namd custom configuration file>
-pb_gb_file              <PB or GB custom configuration file>
-pb_gb_temperature       <temperature> -- default: 310K
-topology_type           <topology filetype> -- default: auto
-trj_type                <trajectory filetype> -- default: auto
-force_parameters        <force field parameters> -- default: auto-detect
-output_filename         <output filename> -- default: mmgbpbsa.log
-debug                   <debug level> -- default: 0. It can be 0, 1, 2. If '2' is selected, it will save all
                         generated temporary files. If '1' is selected, it will save some of the generated
                         temporary files. If '0' is selected, it will only save the result files.
-first_frame             <first frame> -- default: 0
-last_frame              <last frame> -- default: -1
```

-frame_stride	<stride> -- default: 1
-complex_selection	<complex selection> -- default: ""
-receptor_selection	<receptor selection> -- default: ""
-ligand_selection	<ligand selection> -- default: ""
-molecular_mechanics	<do gas-phase calc> -- default: 0
-namd_executable	<path to NAMD> -- default: "namd2"
-namd_cutoff	<cutoff for MD simulations> -- default: 16.0
-namd_switching	<"on" means smooth switching function truncates van der Waals potential energy at the cutoff distance. "off" will close smooth switching function> -- default: on
-namd_switchdist	<distance at which to activate switching function> -- default: "namd2"
-mm_dielectric	<dielectric constant> -- default: 1.0
-poisson_boltzmann	<do PB calculation> -- default: 2. It can be '0, 1, 2'. If '0' is selected, it will calculate MM/GBSA. If '1' is selected, it will calculate MM/PBSA with the program Delphi. If '2' is selected, it will calculate MM/PBSA with the program APBS.
-pb_executable	<path to DelPhi/APBS> -- default: "apbs"
-pb_radii_type	<type of PB radii> -- default: bondi
-pb_boundary_flag	<boundary condition> -- default: sdh
-pb_charge_method	<charge method> -- default: spl0
-ion_charge_apbs_negtive	<total anionic charge> -- default: -1.0
-ion_charge_apbs_positive	<total cationic charge> -- default: 1.0
-ion_apbs_radius	<mobile ion species radius> -- default: spl0
-generalized_born	<do GB calculation> -- default: 0
-gb_exterior_dielectric	<external dielectric> -- default: 78.5
-pb_gb_ion_concentration	<ion concentration> -- default: 0.15
-gb_surface_area	<do LCPO calculation> -- default: 0
-gb_surface_gamma	<surface tension> -- default: 0.0072
-surface_accessibility	<do SA calculation> -- default: 1, it can be 1 or 2, "if '1', SASA calculation is performed by VMD; if '2', SASA calculation is performed by APBS.
-sa_input_apbs	<APBS input file for SA calculation> -- default: ""
-sa_radii_type	<type of SA radii> -- default: bondi
-sa_gamma	<surface tension> -- default: 0.0072
-sa_beta	<surface offset> -- default: 0.0
-sa_probe_radius	<radius of probe> -- default: 1.4
-sa_displacement_apbs	<displacement parameter for finite-difference-based force calculations (surface area derivatives), with unit: Å.> -- default: 0.2
-sa_sample_points	<number of samples> -- default: 500
-namd_seed	<namd seed for reproducibility> -- default: 12345
-num_calc_cores	<number of CPU cores for calculations> -- default: 1

Part III. Decomposing the free energy contributions of a per-residue

3.2. Decomposing the free energy

If the users want to use MolAICal to decompose the free energy contributions of a per-residue, it can draw lessons from the above steps. For example, the residue M165 of SARS-CoV-2 Mpro is reported to interact with ligand N3. So the residue M165 is chosen as an example. First of all, change the console into the “020-mmpbsa\Decompose” folder:

#> cd 020-mmpbsa\Decompose

The calculation script and method for decomposing the free energy contributions are identical to the example above. The only difference is the precise selection of residues for residue energy decomposition and the ligand molecule, which requires modifying the following parameters:

```
-complex_selection        "protein and resid 165 or resname LIG" \
-receptor_selection       "protein and resid 165"   \
-ligand_selection         "resname LIG" \
```

Note: "-complex_selection", "-receptor_selection", and "-ligand_selection" are used to select the molecular part of the complex, receptor, and ligand that is like the command "atomselect" in the VMD Tk console.

And run the below command that can quickly calculate residue free energy contributions based on MM/PBSA:

#> molaical.exe -mmpbgbsa -f dec_mmpbsa_apbs.vmd

The output contains the binding free energy of a per-residue △G as follows:

Delta:					
BOND:	25	-0.0000		0.0000	0.0001
ANGLE:	25	-0.0000		0.0000	0.0000
DIHED:	25	0.0000		0.0000	0.0000
IMPRP:	25	0.0000		0.0000	0.0000
KINETIC:	25	7.2977		0.0000	0.0000
Elec:	25	-2.2492		0.1908	0.9541
Vdw:	25	-5.3395		0.2359	1.1793
PB:	25	1.6785		1.9886	9.9430
SA:	25	-1.2774		0.0397	0.1984
Gas:	25	-7.5887		0.3867	1.9333
Sol:	25	0.4011		1.9851	9.9256
Pol:	25	-0.5707		1.9595	9.7977
Npol:	25	-6.6169		0.2676	1.3378
G binding:	**25**	**-7.1876**	+/-	**1.9791**	**9.8957**

* All energy values are in kcal/mol

Note:

✧ The users can use the above similar method to calculate the free energy contribution of pairwise and per-residue by MolAICal. Results may vary across different operating systems or NAMD versions, but if two decimal places are retained, the results will be identical or very similar.

✧ It has a very similar way for “Input files and result analysis” as the above results. Please check them.

Generally speaking, MolAICal supplies a fast way to **calculate MM/PBSA** that only needs to input **one command if the parameters are preset appropriately**.

Part IV. Entropy calculation

3.3. Entropy calculation by Carma and MolAICal

3.3.1. Download the Carma

Carma can be run on Windows or Linux operating systems. The new version of Carma can be downloaded from https://github.com/glykos/carma, https://github.com/glykos/carma, http://utopia.duth.gr/~glykos/progs, or http://utopia.duth.gr/~glykos.

I recommend the latest version of Carma for entropy calculation. The previous version will show the “NaN” errors, etc. Please check the issue report below:
https://groups.google.com/g/carma-molecular-dynamics/c/KpyY5sEkrj4

Here, 25 frames are extracted from DCD trajectories for entropy calculation. The users can extract a suitable number of frames for their projects. More number of frames will cost more running time.

According to the LICENSE of Carma (https://github.com/glykos/carma/blob/master/LICENSE), which is without the limitation rights to use, copy, modify, merge, publish, distribute, etc. Hence, MolAICal can directly integrate and invoke Carma to fit the molecules and calculate the entropy of molecules **without downloading Carma by users again.**

Option: here, the input files such as 'mpro.dcd' and 'mpro.psf' consist of protein, ligand, ions, water, etc, which may occupy much computer memory. The entropy calculations may require much time and memory. To **save memory and improve computation speed**, users can use the command below to only extract the trajectory of complex (receptor and ligand), receptor, and ligand, respectively:

1. For complex (1st molecule):

```
#> molaical.exe -call run -n vmd -c stripDCD -i -dispdev text -psf "mpro.psf" -args protein,or,resname,LIG "mpro.dcd" "complex" mpro.psf mpro.pdb
```

Note: "complex" is the prefix of the output files. The mpro.psf and mpro.pdb are referred to generate the files that have the prefix "complex". It will generate 'complex.psf', 'complex.pdb', and 'complex.dcd'.

2. For receptor (2nd molecule):

```
#> molaical.exe -call run -n vmd -c stripDCD -i -dispdev text -psf "mpro.psf" -args protein "mpro.dcd" "protein" mpro.psf mpro.pdb
```

Note: "protein" is the prefix of the output files. The mpro.psf and mpro.pdb are referred to generate the files that have the prefix "protein". It will generate 'protein.psf', 'protein.pdb', and 'protein.dcd'.

3. For ligand (3rd molecule):

```
#> molaical.exe -call run -n vmd -c stripdcd -i -dispdev text -psf "mpro.psf" -args resname,LIG "mpro.dcd" "ligand" mpro.psf mpro.pdb
```

Note: "ligand" is the prefix of the output files. The mpro.psf and mpro.pdb are referred to generate the files that have the prefix "ligand". It will generate 'ligand.psf', 'ligand.pdb', and 'ligand.dcd'.

3.3.2. Calculate the complex entropy

#> cd 020-mmpbsa\entropy

Go to the folder “com”. This step is for removing rotations and translations.

#> molaical.exe -call run -c entropy -i -v -fit -force -atmid ALLID -segid A -segid C complex.dcd complex.psf

Note: the name of the generated file "carma.fitted.dcd" is fixed, and the "carma.fitted.dcd" only contains the corresponding contents of "-segid". Users can optimize and adjust the commands for their research according to this prompt.

-call:	when its value is "run", it will call the external program.
-c:	the calculation way, it will be used to calculate entropy by Carma if its value is "entropy".
-i:	after "-i", follow with Carma-related commands. The command-line argument "-i" **must** be placed **after** other arguments (e.g., after argument "-c").
-v:	verbose information
-fit:	Remove global rotations-translations from DCD frames
-force:	tells Carma to stop the calculation of entropies before they become undefined.
-atmid:	atom-name-based atom selection
-segid:	segment-identifier-based atom selection (dcd-psf only)

This step is for calculating entropy.

#> molaical.exe -call run -c entropy -i -v -cov -eigen -mass -force -temp 310 -atmid ALLID -segid A -segid C -last 25 carma.fitted.dcd complex.psf >& com.log &

-call:	when its value is "run", it will call the external program.
-c:	the calculation way, it will be used to calculate entropy by Carma if its value is "entropy".
-i:	after "-i", follow with Carma-related commands. The command-line argument "-i" **must** be placed

	after other arguments (e.g., after argument "-c").
-v:	verbose information
-cov:	calculate variance-covariance matrix
-eigen:	calculate eigenvectors and values of covariance matrices
-mass:	use mass-weighting for the covariance matrix
-force:	tells carma to stop the calculation of entropies before they become undefined.
-atmid:	atom-name-based atom selection
-segid:	segment-identifier-based atom selection (dcd-psf only)

3.3.3. Calculate the receptor entropy

Go to the folder “rec”. This step is for removing rotations and translations.
#> molaical.exe -call run -c entropy -i -v -fit -atmid ALLID -segid A protein.dcd protein.psf

This step is for calculating entropy.
#> molaical.exe -call run -c entropy -i -v -cov -eigen -mass -force -temp 310 -atmid ALLID -segid A -last 25 carma.fitted.dcd protein.psf >& rec.log &

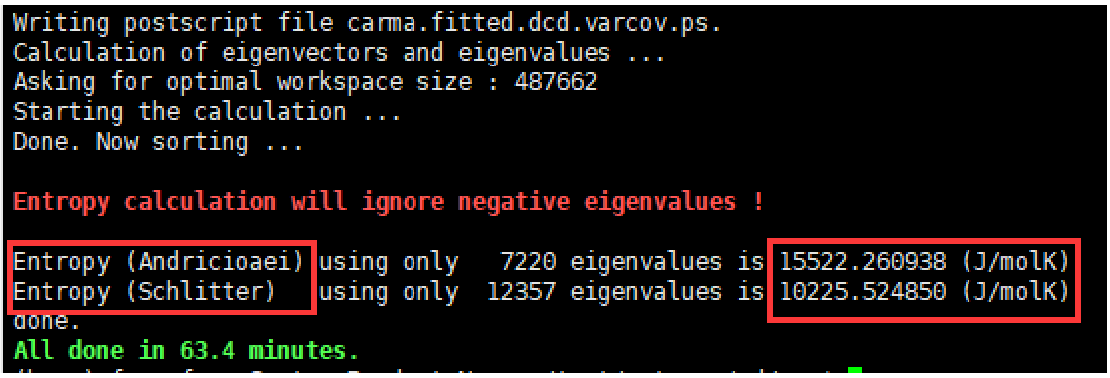
3.3.4. Calculate the ligand entropy

Go to the folder “lig”. This step is for removing rotations and translations.
#> molaical.exe -call run -c entropy -i -v -fit -force -atmid ALLID -segid C ligand.dcd ligand.psf

This step is for calculating entropy.
#> molaical.exe -call run -c entropy -i -v -cov -eigen -mass -force -temp 310 -atmid ALLID -segid C -last 25 carma.fitted.dcd ligand.psf >& lig.log &

3.3.5. Calculate temperature multiplied by the delta entropy

Please check log files: “com.log”, “rec.log”, and “lig.log”, and find the Andricioaei’s or Schlitter’s entropy values in the files of “com.log”, “rec.log”, and “lig.log”. For example, it can find the Andricioaei’s or Schlitter’s entropy value in the content of “com.log” (see the figure below):



Change to the parent directory, run MolAICal based on log files: “com.log”, “rec.log”, and “lig.log” as follows:

```
#> molaical.exe -entropy -if -c com/com.log -r rec/rec.log -l lig/lig.log -t 310.0
```

It will show the result:

```
Entropy Type: Andricioaei
The entropy (T * delta S) = -25.3034 (kcal/mol)

Entropy Type: Schlitter
The entropy (T * delta S) = -32.3195 (kcal/mol)
```

> Where: **-entropy:** means delta entropy calculation
> **-c:** the value or file of complex entropy. Here, it is a result file.
> **-r:** the value or file of the receptor or first molecule entropy. Here, it is a result file.
> **-l:** the value or file of the ligand or second molecule entropy. Here, it is a result file.
> **-t:** the Kelvin temperature in MD simulation.
> **if:** it will read entropy values from result files, if the input arguments contain "-if".

Note: Calculating entropy under Windows operating system is very slow, possibly only reaching Schlitter's entropy value; it is recommended to perform entropy calculations on a Linux operating system instead.

If entropy is considered, the MM/PBSA is estimated by the following equations:

$\Delta G_{\text{bind}} = \Delta H - T\Delta S \approx \Delta E_{\text{MM}} + \Delta G_{\text{sol}} - T\Delta S$

$\Delta E_{\text{MM}} = \Delta E_{\text{internal}} + \Delta E_{\text{ele}} + \Delta E_{\text{vdw}}$

$\Delta G_{\text{sol}} = \Delta G_{\text{PB}} + \Delta G_{\text{SA}}$

$\Delta G_{\text{SA}} = \gamma * \text{SASA} + \beta$

As mentioned above, the energy value of entropy was not considered, as follows:
delta G binding: -66.5006 +/- 2.7810 (kcal/mol)

Using Entropy Type: Andricioaei
ΔG_{bind} = -66.5006 – (-25.3034) = -41.1972 (kcal/mol)

Using Entropy Type: Schlitter
ΔG_{bind} = -66.5006 – (-32.3195) = -34.1811 (kcal/mol)

Note: If there are no binding-induced structural changes in the process of molecular dynamics (MD) simulations, or very similar structural changes among different systems, the entropy calculation can be omitted. Using only the △G binding value without considering entropy is sufficient.

Options:

✧ **MolAICal also provides a simple way for the entropy calculation as follows:**

#> molaical.exe -call run -c script -n runscript -i com rec lig

The strings after '-i' are the parameter list that will be split by space. The split strings will replace the strings $molargs1, $molargs2, $molargs3, etc. in the script file. The number of the string $molargs1, $molargs2, $molargs3, etc., is the same as the length of the split strings after '-i'. The **'-n'** assigns the name of the script file named "runscript" as follows:

```
# 1. calculate complex part (rename in win: move /y carma.fitted.dcd complex_fit.dcd)
cd $molargs1
molaical.exe -call run -c entropy -i -v -fit -force -atmid ALLID -segid A -segid C complex.dcd complex.psf
molaical.exe -call run -c entropy -i -v -cov -eigen -mass -force -temp 310 -atmid ALLID -segid A -segid C -last 25 carma.fitted.dcd complex.psf > com.log
cd ..

# 2. calculate receptor part
cd $molargs2
molaical.exe -call run -c entropy -i -v -fit -atmid ALLID -segid A protein.dcd protein.psf
molaical.exe -call run -c entropy -i -v -cov -eigen -mass -force -temp 310 -atmid ALLID -segid A -last 25 carma.fitted.dcd protein.psf > rec.log
cd ..

# 3. calculate receptor part
cd $molargs3
molaical.exe -call run -c entropy -i -v -fit -force -atmid ALLID -segid C ligand.dcd ligand.psf
molaical.exe -call run -c entropy -i -v -cov -eigen -mass -force -temp 310 -atmid ALLID -segid C -last 25 carma.fitted.dcd ligand.psf > lig.log
cd ..

# in windows: | findstr /I /R "entropy" > results.log     in linux: | grep -i "entropy" > results.log
molaical.exe -entropy -if -c $molargs1/com.log -r $molargs2/rec.log -l $molargs3/lig.log -t 310.0 > results.log
```

Note: "molaical.exe" does not need to be modified; MolAICal will automatically replace its path for "molaical.exe".

Appendix 1

Install external programs inside the MolAICal container (if finished, only for Linux Version MolAICal)

Please note that the configuration of MolAICal differs between the Windows and Linux versions: the Linux version utilizes the udocker container approach and requires setup within the container, whereas the Windows version does not.

To address the library dependency issues in Linux, the Linux version of MolAICal adopts a container-based approach. If **external programs need** to be invoked, it is **recommended** to install them **within** the MolAICal **container**. If **no external** programs are required, this step can be **ignored**. This tutorial requires the external programs NAMD and VMD, and the steps are as follows:

a) First of all, copy the files to the MolAICal container.

```
****** 1.    Go to the container file system, it will enter the "/root" ******
#> molaical.exe -eset shell in

****** 2.    The first part of 'cp' is from the local machine, and the second part is in the container; ******
****** The VMD and NAMD packages can be moved into the container by the command below ******
#> cp /home/user/<local machine file> /root/soft

****** 3.    Exit container file system ******
#> exit
```

b) Secondly, enter the container virtual environment of MolAICal:

```
#> molaical.exe -eset sys run molaical
```

Note: 'molaical' is the container name.

c) Installing software in the container virtual environment of MolAICal is the same as installing it on the host local machine. Here, we take the installation of VMD and NAMD as examples:

1) For NAMD:

Extract the NAMD files, assuming the extracted folder is named namdcpu. Then specify its path to MolAICal using the following command:

```
#> molaical.exe -call set -n NAMD -p "/root/soft/namdcpu/namd3"
```

Note: Please **replace** the above paths for VMD and NAMD with the **actual paths** on the users' system. The string "VMD" and "NAMD" (case insensitive) after "-n" is fixed for the paths of VMD and NAMD. To ensure the reproducibility of MM/GBSA results, it is recommended to use the CPU version of NAMD, as the "seed" parameter appears to be ineffective for reproducibility in the CUDA version of NAMD.

2) For VMD:

Following the steps below:

✧ Extract the VMD files:

```
#> tar -xzvf vmd-xxx.tar.gz
```

Note: Please **replace** the above paths for VMD and NAMD with the **actual paths** on the users' system.

✧ Modify the install path in a file named "**configure**" in the decompressed folder of VMD:

```
The default: $install_bin_dir="/usr/local/bin"; modify it to →
$install_bin_dir="/root/soft/vmd193/bin";

The default: $install_library_dir="/usr/local/lib/$install_name"; modify it to →
$install_library_dir="/root/soft/vmd193/lib/$install_name";
```

✧ Install VMD:

```
#> cd vmd-xx
#> ./configure LINUXAMD64
#> cd src
#> make install
```

Note: Please run **./configure**, and select the right types for the used computers. Here, it is "LINUXAMD64".

✧ Then specify its path to MolAICal using the following command:

```
#> molaical.exe -call set -n VMD -p "/root/soft/vmd193/bin/vmd"
```

So far, all the installations and configurations of NAMD and VMD are complete in the MolAICal container.

✧ **Remember to use the "exit" command to quit the MolAICal virtual environment and return to the local computer for computation (primarily to avoid file-copying steps; execution within the container is also possible but requires copying data from the local computer to the MolAICal container).**

```
#> exit
```

3) Save and load the modified container (Option)

To preserve changes made within a container and avoid reinstalling software later, since all data will be lost if the container is deleted.

◆ The first step is to obtain the container ID and name:

```
#> molaical.exe -eset sys ps
```

◆ Secondly, clone this container as follows:

```
#> molaical.exe -eset sys export --clone -o molaicalv2.tar molaical
```

Note: 'molaical' is the container name. It can also be the container ID. If an error is reported, it may be due to permission issues with the mounted file system, which can be ignored.

- Thirdly, import the cloned container **if needing**:

```
#> molaical.exe -eset sys import --clone --name molaicalv2 molaicalv2.tar
```

- Now, change the loaded new container name to the default container name "molaical", the previous container is renamed to "bakmolaical", the commands are as follows:

```
#> molaical.exe -eset sys rename molaical bakmolaical
#>molaical.exe -eset sys rename molaicalv2 molaical
```

Notice: Please note that a **container** (**dynamic**) is generated from an **image** (**static**). Operations such as software installation must be performed within the dynamic container; if cloning this container, restoring it will still be based on the original underlying image.
For more information, please see the manual of MolAICal or run "molaical.exe -eset sys --help" to get more help details.